

# Ain Foundry Virtual Entrepreneur-in-Residence

Capstone Proposal, Research Design, Scope, and Acceptance Plan



|                                      |                                    |                       |
|--------------------------------------|------------------------------------|-----------------------|
| <b>~120</b>                          | <b>7</b>                           | <b>24/7</b>           |
| venture concepts / planning baseline | EIR profiles currently listed [S3] | target founder access |

|                        |                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------|
| <b>Prepared by</b>     | Satvik Sathish Adyanthaya                                                                          |
| <b>Project sponsor</b> | Professor Roberto Colangelo, Executive Director, Ain Center for Entrepreneurship and Innovation    |
| <b>Audience</b>        | Professor Roberto Colangelo, capstone faculty, Ain Center leadership, and the student project team |
| <b>Date</b>            | September 2026                                                                                     |

**Working name:** Ain Foundry Virtual Entrepreneur-in-Residence (vEIR). This document proposes a human-led, AI-amplified capability; it does not authorize deployment or the use of any person's likeness.

DECISION FRAME

1. Project charter

Purpose

Determine whether a persistent, evidence-grounded virtual Entrepreneur-in-Residence can expand access to high-quality venture coaching while protecting founder confidentiality, intellectual property, and the authority of human EIRs.

Sponsor-defined problem

The Ain Center supports ventures spanning undergraduate consumer concepts, software, advanced manufacturing, biomedical research, and PhD-led commercialization. The sponsor briefing described experienced EIRs as scarce and costly relative to the breadth and number of ventures requiring guidance. The proposed system is therefore a **venture-building operating system, not a generic chatbot**: it must remember each venture, diagnose its situation, challenge weak assumptions, teach, advise, and route founders to the right human or external resource. [S1]

|                                                  |                                                   |                                                |
|--------------------------------------------------|---------------------------------------------------|------------------------------------------------|
| 116                                              | ~120                                              | 1 per venture                                  |
| concepts reported in the last academic year [S2] | active planning scale stated in the briefing [S1] | persistent virtual EIR assignment concept [S1] |

Primary research question

Can a consented, EIR-specific AI avatar with persistent venture memory provide timely, rigorous, and context-aware guidance that founders and EIRs judge useful, while reliably grounding claims, preventing cross-venture disclosure, and escalating consequential questions?

Capstone decision at completion

- **Proceed:** the prototype meets the minimum safety and quality gates and merits a controlled Ain Foundry pilot.
- **Revise:** the venture-memory and advisory core is promising, but likeness, latency, quality, or governance requires another build cycle.
- **Stop:** the system cannot demonstrate acceptable privacy isolation, consent control, grounding, or human oversight.

EVIDENCE BEFORE CAPACITY ASSUMPTIONS

1A. Official EIR baseline

The official Ain Center directory currently lists **seven** Entrepreneurs-in-Residence. It describes an EIR as a seasoned professional who guides founders, refines business ideas, and helps navigate venture challenges. The directory is a public profile count; it does not establish current availability, Foundry assignments, consent to a virtual persona, or equal capacity. Those facts must be confirmed in discovery. [S3]

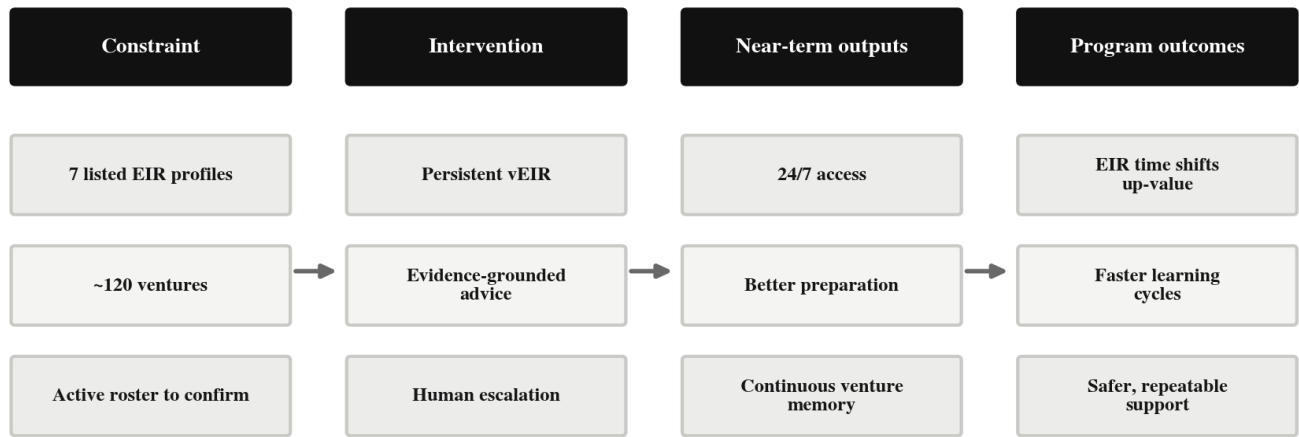
| Listed EIR          | Publicly described focus                                                       | Initial discovery / routing signal                                                                |
|---------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| John Schloff        | Growth, marketing leadership, product management, emerging technology          | Go-to-market design; customer process; product portfolio; traditional-market entry                |
| David Mammano       | Entrepreneurship, leadership, education, motivation                            | Founder leadership; narrative; workshop-style coaching; resilience and communication              |
| Kate Washington     | Finance, operations, economic development, global growth                       | Operating model; financial discipline; public-private partnerships; international expansion       |
| Roberto Colangelo   | Strategy, entrepreneurship, global marketing, biotech, energy                  | Venture strategy; commercialization; emerging markets; fundraising and government modernization   |
| Scott Walker        | Finance, operations, capital projects, sales and investor relations            | Startup finance; operating cadence; capital planning; sales execution; investor readiness         |
| Dina Uzri           | Science-to-market commercialization, biotech, product and business development | Research translation; value proposition; life-science pathway; fundraising and strategic planning |
| Truptesh H. Kothari | Clinical program building, health-system operations, multidisciplinary growth  | Clinical innovation; healthcare delivery; stakeholder alignment; quality and scalable programs    |

Capstone implication

Prototype one consenting EIR persona first. Treat expertise matching as a governed routing recommendation, not an automatic assignment, and validate every profile with the EIR before using it as doctrine.

THEORY OF CHANGE

2. Problem-to-outcome logic



Human EIRs remain accountable for consequential guidance and program decisions.

Figure 1. Sponsor need translated into a measurable intervention and program outcomes. Counts are planning baselines, not audited capacity data.

Value hypothesis

Routine guidance, preparation, and continuity can be delivered safely by a virtual EIR; scarce human time can then concentrate on judgment-heavy moments such as IP strategy, regulated pathways, financing, founder conflict, and high-stakes external commitments. The system succeeds only if it improves the quality of human engagement rather than merely increasing message volume.

|                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Current-state friction</b></p> <p>Founder context is repeatedly reconstructed; advice availability depends on calendars; EIR capacity is uneven across disciplines; and promising issues can surface too late.</p> | <p><b>Target-state behavior</b></p> <p>Every venture has a persistent workspace; founders can practice and diagnose at any time; the system explains its evidence and uncertainty; and the right human receives a structured escalation packet.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Non-negotiable design principles

- **Amplify, never impersonate:** the interface must continuously disclose that it is synthetic and that the human EIR did not personally author each response.
- **Persistent but correctable:** memory is visible, source-linked, editable, and never treated as infallible.
- **Tough but constructive:** skepticism means testing evidence and exposing assumptions, not hostility or personality theater.
- **Venture-private by default:** raw venture content never becomes another venture's retrieval context.
- **Human authority:** the EIR and Ain team define doctrine, approve the persona, review escalations, and can suspend the avatar immediately.

## BOUNDARIES

### 3. Objectives, scope, and stakeholders

| Objective             | Capstone result                                                                                                    | Evidence of completion                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| O1 - Requirements     | Translate sponsor expectations into testable functional, safety, data, and operating requirements.                 | Approved backlog and traceability matrix.                         |
| O2 - vEIR core        | Build one real-time or near-real-time conversational prototype with one EIR persona and persistent venture memory. | Demonstrable founder sessions across multiple venture workspaces. |
| O3 - Advisory quality | Implement diagnosis, challenge, coaching, resource routing, uncertainty, citations, and escalation.                | Blind EIR rubric and scenario evaluation.                         |
| O4 - Privacy          | Enforce venture-scoped retrieval and test against unauthorized cross-venture access.                               | Red-team suite with zero critical leakage.                        |
| O5 - Handoff          | Deliver deployable design, source, runbook, backlog, and pilot recommendation.                                     | Reproducible build and sponsor decision memo.                     |

#### In scope

- One consented EIR persona or, until approvals are complete, a clearly non-identifiable surrogate body connected to the intended EIR knowledge and style pipeline.
- Founder onboarding, venture profile, persistent memory, document-grounded question answering, diagnostic coaching, pitch or decision rehearsal, action recap, and escalation.
- A pilot dataset of EIR-approved material and synthetic or expressly authorized venture cases.
- A browser experience accessible to selected participants with role-based authentication and audit logs.
- Evaluation of utility, persona fidelity, grounding, continuity, privacy, latency, and escalation behavior.

#### Out of scope for the capstone

- Autonomous legal, securities, tax, medical, regulatory, IP-ownership, or investment decisions.
- Submitting applications, sending external communications, signing documents, moving funds, or representing that the human EIR approved an action.
- Full deployment to all 120 ventures, all EIRs, or restricted laboratory data before institutional review.
- Training a foundation model from scratch or fine-tuning on raw confidential venture conversations.

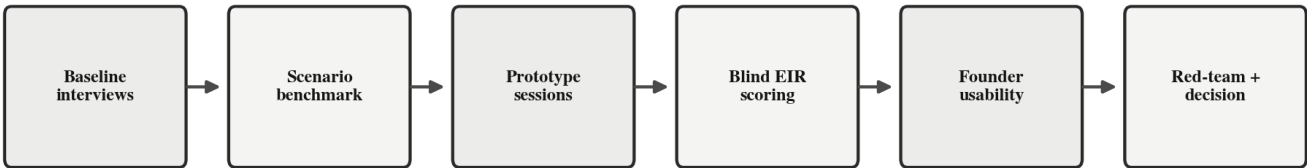
#### Stakeholders

| Stakeholder     | Role                                                              | What must be protected                                  |
|-----------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Founders        | Use the vEIR, correct memory, decide whether to escalate.         | Venture IP, candor, autonomy, and clear disclosure.     |
| Human EIRs      | Define doctrine, approve likeness and voice, calibrate advice.    | Reputation, likeness rights, time, and final authority. |
| Ain team        | Own program policy, access, routing, and pilot operations.        | Program trust, consistency, and manageable workload.    |
| UR stakeholders | Review security, privacy, research, contracts, and accessibility. | Institutional policy, FERPA, vendor terms, and risk.    |
| Capstone team   | Design, build, test, document, and hand off.                      | Clear scope, data access, and reproducibility.          |

METHOD

4. Research and evaluation design

The recommended capstone method is a stage-gated mixed-method evaluation. It tests the advisory core before treating visual realism as proof of usefulness. The smallest credible cohort is one EIR and 8-12 ventures, with at least three structured interactions per venture over four weeks. This is a proposed design; recruitment and any research-review determination remain sponsor decisions.



One evidence chain from need to recommendation

Figure 2. Evaluation sequence. A functioning visual avatar is not sufficient without quality, privacy, and escalation evidence.

Study components

| Component          | Participants / data                                      | Measure                                                                               |
|--------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Discovery          | Sponsor, Ain operators, participating EIR, 5-8 founders. | Critical jobs, failure modes, data boundaries, current wait and preparation friction. |
| Scenario benchmark | 30-50 representative prompts across stages and domains.  | Expert reference response, required evidence, escalation label, prohibited content.   |
| Prototype pilot    | 8-12 venture teams; three or more sessions each.         | Completion, latency, repeat-context burden, citation use, next-action adoption.       |
| Expert review      | Participating EIR blind-rates sampled responses.         | Correctness, rigor, specificity, teachability, persona fit, and escalation.           |
| Privacy red team   | At least 100 access-control and prompt-injection cases.  | Unauthorized retrieval, memorization, data exfiltration, policy bypass.               |
| Post-pilot         | Founder survey and interviews; EIR debrief.              | Trust, utility, over-reliance, workload shift, and readiness to continue.             |

Analysis plan

- Report medians and distributions, not only averages, for latency, scores, and usage.
- Separate factual correctness from coaching quality and from visual or voice fidelity.
- Code qualitative feedback into access, continuity, rigor, trust, usability, and safety themes.
- Compare first-session and later-session ratings to test whether persistent context produces measurable improvement.
- If findings will be generalized or published, obtain an institutional determination on human-subjects review before data collection.

FUNCTIONAL SPECIFICATION

5. Prototype behavior and requirements

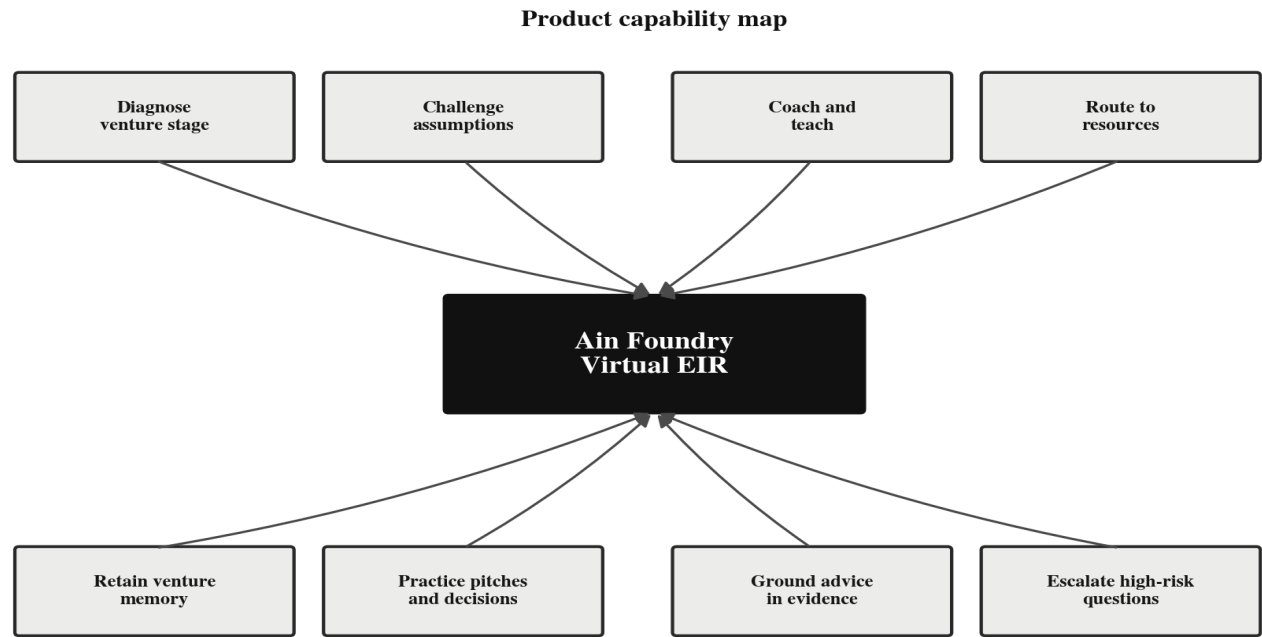


Figure 3. Capstone capability boundary derived from the sponsor briefing.

| ID   | Requirement                                                                                         | Pilot verification                        |
|------|-----------------------------------------------------------------------------------------------------|-------------------------------------------|
| F-01 | Assign one vEIR and one isolated memory workspace to each pilot venture.                            | Access and workspace tests.               |
| F-02 | Retrieve approved venture facts, artifacts, decisions, and milestones with source attribution.      | Memory benchmark and citation inspection. |
| F-03 | Diagnose venture stage, expose assumptions, and ask evidence-seeking follow-ups before prescribing. | Scenario rubric.                          |
| F-04 | Shift between coaching, teaching, rehearsal, and resource-routing modes without losing context.     | Scripted multi-turn sessions.             |
| F-05 | Express uncertainty and distinguish fact, inference, hypothesis, and recommendation.                | Calibration test.                         |
| F-06 | Generate a founder-visible recap with decisions, risks, open questions, and next actions.           | Session artifact review.                  |
| F-07 | Escalate high-risk topics with a concise packet for a human EIR or specialist.                      | Escalation test set.                      |
| F-08 | Allow founder and EIR to inspect, correct, pin, or delete eligible memory entries.                  | UI and audit-log test.                    |
| F-09 | Disclose synthetic identity throughout the session and on exported artifacts.                       | Interface inspection.                     |
| F-10 | Support immediate persona suspension and access revocation.                                         | Administrative control test.              |

TRUST BEFORE REALISM

6. Data, consent, and governance

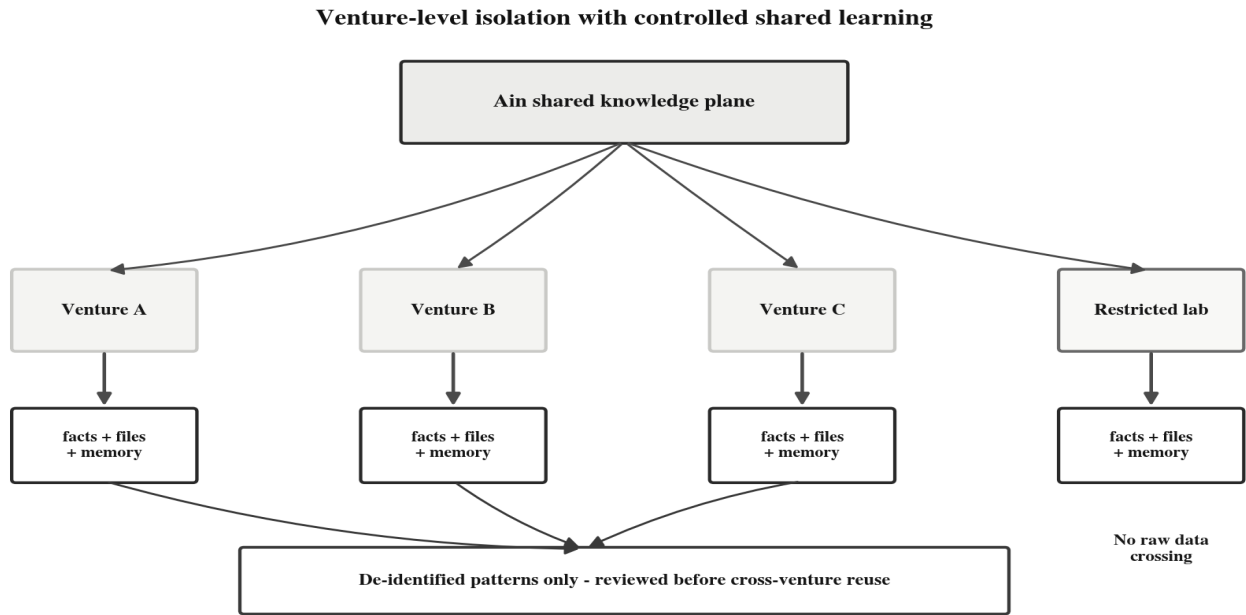


Figure 4. Raw venture data remains inside its authorized workspace; only reviewed, de-identified program patterns may cross venture boundaries.

Data classes and capstone treatment

| Class                  | Examples                                                                   | Capstone rule                                                                       |
|------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Public                 | Ain pages, public curricula, public market or regulatory resources.        | Allowed with provenance and refresh date.                                           |
| Program internal       | Ain playbooks, referral lists, rubrics, internal schedules.                | Use only after owner approval and access control.                                   |
| EIR persona            | Video, voice, cases, critique style, doctrine, consent records.            | Separate asset store; explicit use scope; EIR approval before release.              |
| Venture private        | Pitch decks, experiments, customers, finances, transcripts, decisions.     | Venture-scoped encryption and retrieval; no model training by default.              |
| Restricted             | Lab IP, patient or clinical data, export-controlled material, legal files. | Excluded from V1 unless the University explicitly approves the system and workflow. |
| De-identified learning | Reviewed patterns stripped of venture and person identifiers.              | Publish to shared library only through an approval workflow.                        |

Consent and identity controls

- Separate written consent for visual likeness, voice, training material, intended audiences, approved use cases, retention, and revocation.
- Keep recorded provider-required consent statements and contractual disclosures in the governance record. Microsoft and HeyGen both impose consent controls for custom avatars. [S5][S8]
- Display persistent synthetic-media labeling; do not allow downloads or public reuse of the avatar unless separately approved.
- Let the EIR review representative sessions, correct doctrine, suspend the system, and retire the persona.
- Route all data handling through University review. UR guidance emphasizes approved tools, data classification, and special care for proprietary, FERPA, health, and other non-public data. [S11][S12][S13]

CONTROL DESIGN

7. Risk, safety, and escalation

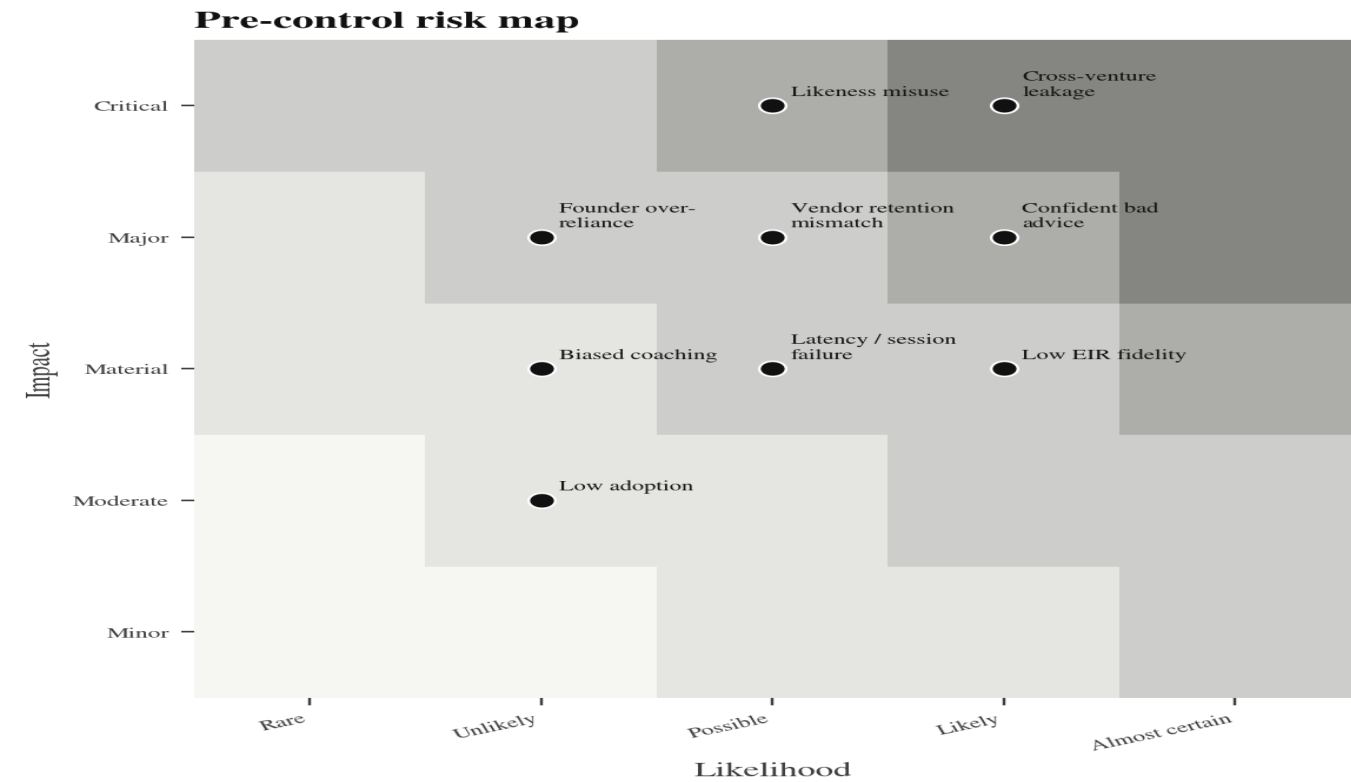


Figure 5. Proposed pre-control risk map. Likelihood and impact must be recalibrated during discovery.

| Trigger                                                       | vEIR response                                                                            | Human destination                                     |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|
| IP ownership, patentability, licensing, lab or sponsor rights | Summarize known context, state limits, stop prescriptive advice, create an issue packet. | UR Ventures or authorized counsel.                    |
| Medical, clinical, FDA, safety, or human-subjects pathway     | Offer general process education only; flag missing evidence and uncertainty.             | Qualified regulatory, clinical, or research contact.  |
| Financing terms, securities, valuation, investor commitments  | Explain concepts and tradeoffs; do not recommend or transmit a commitment.               | Human EIR, finance specialist, or counsel.            |
| Founder conflict, harassment, crisis, or wellbeing            | Avoid mediation theater; encourage immediate human support.                              | Ain program lead and appropriate University resource. |
| Low grounding, conflicting sources, or high uncertainty       | Say what is unknown, request evidence, and withhold confident direction.                 | Human EIR or domain specialist.                       |
| Attempted cross-venture access or prompt injection            | Deny, end affected tool path, record security event.                                     | System administrator and security reviewer.           |

Safety gates

- No critical cross-venture leakage in the capstone red-team suite.
- All high-risk scenarios produce an appropriate limitation and escalation path.
- No session starts for a revoked persona, unauthorized user, or mismatched venture scope.
- Every substantive answer can expose its sources, venture memory entries, and uncertainty.
- Ain and the EIR can suspend the system without engineering intervention.

SEPTEMBER-DECEMBER 2026

## 8. Delivery plan and accountability

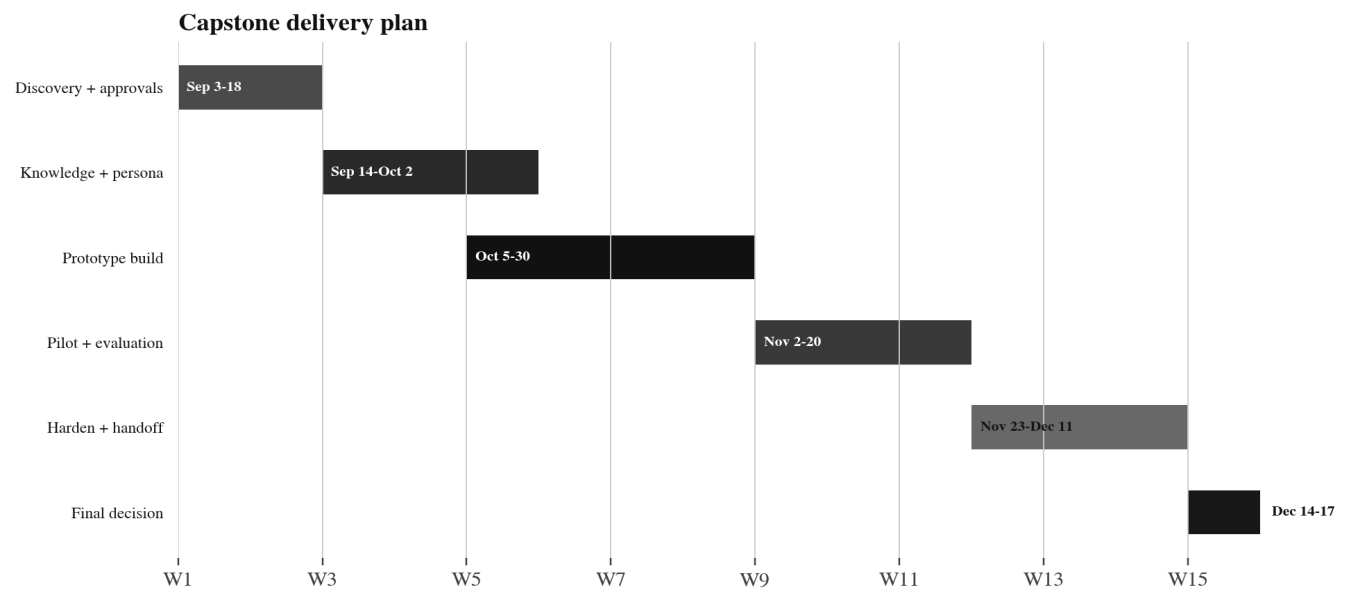


Figure 6. Proposed capstone sequence through December 17, 2026. Dates are management targets and should be aligned to the official course calendar.

| Phase                   | Exit deliverable                                                                        | Decision owner                                 |
|-------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------|
| Discovery and approvals | Charter, stakeholder interviews, data inventory, consent path, and risk classification. | Sponsor plus appropriate University reviewers. |
| Knowledge and persona   | Approved source corpus, EIR doctrine interview, scenario benchmark, and persona card.   | Participating EIR and Ain program lead.        |
| Prototype build         | Authenticated multi-venture app, memory, evidence, escalation, and avatar adapter.      | Capstone team.                                 |
| Pilot and evaluation    | Usage evidence, expert scores, founder feedback, privacy and adversarial test results.  | Capstone team with EIR adjudication.           |
| Hardening and handoff   | Runbook, architecture, backlog, known limitations, and reproducible deployment package. | Capstone team and technical reviewer.          |
| Final decision          | Proceed, revise, or stop recommendation with evidence and a Foundry pilot plan.         | Professor Colangelo and Ain leadership.        |

### RACI summary

| Workstream                                     | Satvik / capstone team | Prof. Colangelo / EIR | Ain / UR reviewers |
|------------------------------------------------|------------------------|-----------------------|--------------------|
| Requirements and prototype                     | R / A                  | C                     | I                  |
| Persona doctrine and likeness                  | R                      | A                     | C                  |
| Founder recruitment and program fit            | C                      | A                     | R                  |
| Security, privacy, vendor, and research review | R                      | C                     | A                  |
| Evaluation and analysis                        | R / A                  | C / adjudicate        | I                  |
| Scale decision                                 | Recommend              | A                     | C                  |

EVIDENCE STANDARD

9. Success metrics and acceptance gates

Targets below are proposed capstone gates, not claims of current performance. Safety gates override average quality scores.

| Dimension        | Metric                                         | Proposed minimum gate              | How tested                                    |
|------------------|------------------------------------------------|------------------------------------|-----------------------------------------------|
| Privacy          | Critical unauthorized disclosure               | 0 events                           | 100+ adversarial and access-control tests.    |
| Continuity       | Approved venture-fact recall                   | >= 90% correct; 100% source-linked | Seeded memory benchmark across sessions.      |
| Grounding        | Material factual claims supported              | >= 90%                             | Expert audit of sampled answers.              |
| Advisory quality | EIR rubric score                               | >= 4.0 / 5 median                  | Blind review across benchmark scenarios.      |
| Escalation       | Recall on high-risk prompts                    | >= 95%                             | Labeled risk test set.                        |
| Calibration      | Unsupported high-confidence statements         | No critical instances              | Conflict and insufficient-evidence scenarios. |
| Experience       | Warm-path first audible response               | P50 <= 800 ms; P95 <= 1.2 s target | Instrumented end-to-end sessions.             |
| Utility          | Founder-reported session value                 | >= 4.0 / 5 median                  | Post-session survey plus interviews.          |
| Human leverage   | EIR says handoff packet reduces reconstruction | >= 80% of reviewed escalations     | EIR debrief and sample comparison.            |

Kill criteria

Any demonstrated cross-venture leakage, uncorrectable likeness or consent defect, systematic unsafe advice, or inability to suspend access blocks deployment regardless of overall satisfaction.

Capstone completeness checklist

- Sponsor-approved charter and traceable requirements.
- Documented institutional and vendor approval status, including unresolved constraints.
- Reproducible prototype with separate venture workspaces and one governed persona.
- Scenario benchmark, expert ratings, founder feedback, latency results, and red-team evidence.
- Architecture, operating blueprint, code, runbook, data dictionary, risk register, and prioritized backlog.
- Final live demonstration and proceed / revise / stop recommendation.

## HANDOFF

## 10. Deliverables, decisions, and source register

### Final capstone package

| Deliverable               | Contents                                                                                           | Acceptance                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| D1 - Product evidence     | Problem definition, user journeys, requirements, prototype, and demonstration script.              | Sponsor can reproduce the key founder journey.                          |
| D2 - Technical package    | Architecture, source, environments, interfaces, data model, security controls, tests, and runbook. | A new technical team can continue without reconstructing design intent. |
| D3 - EIR package          | Consent record, persona card, doctrine, approved sources, scenario rubric, and calibration log.    | EIR approves representative behavior or documents exceptions.           |
| D4 - Evaluation report    | Methods, quantitative results, qualitative findings, failures, limitations, and gate assessment.   | All claims trace to captured evidence.                                  |
| D5 - Scale recommendation | Pilot size, governance, vendor route, cost model, ownership, and next 90-day plan.                 | Ain leadership can make a funded go / revise / stop decision.           |

### Open decisions to resolve in discovery

1. Name the first EIR persona and document the exact approved uses of likeness, voice, style, and expertise.
2. Confirm the authoritative number of active EIRs, current availability, and venture assignments. The official Ain page currently lists seven EIR profiles, while the sponsor briefing referenced a smaller active group and approximately 120 ventures. Treat seven as a public-directory count—not an audited capacity number. [S1][S3]
3. Select approved hosting, model, speech, and avatar vendors after University security, privacy, accessibility, procurement, and contract review.
4. Determine whether pilot activity is program evaluation, class work, or research requiring additional review before recruitment.
5. Define which restricted data classes are excluded, which human roles receive escalations, and how long each data type is retained.

### Core sources

S1 Ain Center for Entrepreneurship and Innovation. Analytics Projects course sponsor briefing with Professor Roberto Colangelo. Video recorded August 31, 2026. User-provided source.

S2 University of Rochester News Center. Student entrepreneurs learn business by starting one. July 29, 2026. <https://www.rochester.edu/newscenter/student-entrepreneurs-become-business-founders-712582/>

S3 Ain Center for Entrepreneurship and Innovation. Entrepreneurs in Residence. Accessed September 2026. <https://www.rochester.edu/aincenter/entrepreneurs-in-residence/>

S5 Microsoft Learn. Limited Access - custom neural voice and custom text to speech avatar. Updated August 26, 2026. <https://learn.microsoft.com/en-us/azure/foundry/responsible-ai/speech-service/text-to-speech/limited-access>

S7 LiveAvatar Documentation. LiveAvatar Overview. Accessed September 2026. <https://docs.liveavatar.com/>

S8 HeyGen Help Center. LiveAvatar FAQ. Updated 2026. <https://help.heygen.com/en/articles/12758866-liveavatar-faq>

S11 University of Rochester. Artificial Intelligence Tools. Accessed September 2026. <https://tech.rochester.edu/artificial-intelligence-ai-tools/>

S12 University of Rochester. Responsible Use of Generative Artificial Intelligence in Research. Accessed September 2026. <https://www.rochester.edu/university-research/about/committees/ai-research-committee/responsible-use-gen-ai/>

S13 University of Rochester. Data Security Classification Policy. Accessed September 2026. <https://www.rochester.edu/policies/policy/data-security-classifications/>

S14 National Institute of Standards and Technology. AI RMF Generative AI Profile, NIST AI 600-1. Updated April 8, 2026. <https://doi.org/10.6028/NIST.AI.600-1>

S15 Project research note supplied by Satvik Sathish Adyanthaya: Azure virtual human, real-time EIR avatars, and HBS Foundry comparison. September 2026. User-provided source; claims independently checked where used.